Breakwater Designer — Methods and References

An integrated conceptual-design tool for rubble-mound and vertical (caisson) breakwaters. This note summarises the published design formulae used in each module and their validity. For conceptual design; detailed design requires physical-model testing.

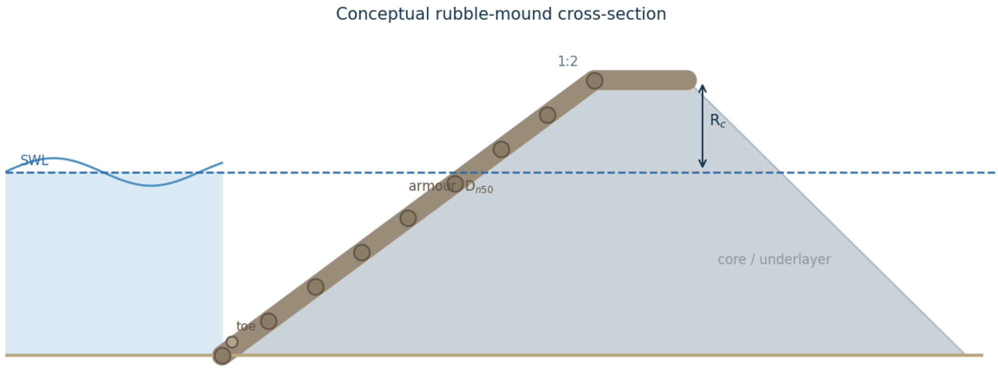

Figure 1. Conceptual rubble-mound cross-section: front slope, primary armour of nominal diameter D_{n50}, core/underlayer, toe berm and crest freeboard R_c.

1. Primary armour stability

Rock and concrete units are sized with Hudson; rock is additionally checked with Van der Meer (1988), and the larger stone governs:

Hudson: $M_{50} = \rho_r H_s^3 / (K_D \cdot \Delta^3 \cdot \cot\alpha)$. Van der Meer (plunging): $H_s/(\Delta D_{n50}) = 6.2 \cdot P^{0.18} (S/\sqrt{N})^{0.2} \xi_m^{-0.5}$.

2. Toe-berm stability

The toe stone is sized from the stability number as a function of the relative toe depth h_t/h and the damage number N_{od} (Rock Manual / Van der Meer):

$H_s/(\Delta D_{n50}) = (2.0 + 6.2 \cdot (h_t/h)^{2.7}) \cdot N_{od}^{0.15}$.

3. Run-up and overtopping

EurOtop (2018) mean formulae give the 2% run-up and the mean overtopping discharge as functions of the Iribarren number ξ, freeboard R_c and the roughness/obliquity factors γ_f, γ_β:

$R_{2\%}/H_s = 1.65 \gamma_f \gamma_\beta \xi$ (capped); $q/\sqrt{(g H_s^3)} = (0.023/\sqrt{\tan\alpha}) \xi \cdot \exp[-(2.7 R_c/(H_s \xi \gamma_f \gamma_\beta))^{1.3}]$.

The rear-side armour is sized relative to the front stone using guidance keyed to the overtopping rate (equal to the front for heavy overtopping, reducing to about half for light overtopping).

4. Wave transmission (low-crested structures)

For low-crested and submerged structures the transmitted wave height follows d'Angremond, Van der Meer & De Jong (1996), with R_c negative when the crest is below still water:

$K_t = -0.4(R_c/H_s) + b \cdot (B/H_s)^{-0.31}(1 - e^{-0.5\xi})$, $0.075 \le K_t \le 0.8$ (b = 0.64 permeable, 0.80 impermeable).

5. Wave reflection

Reflection from the sloping front is estimated with Zanuttigh & Van der Meer (2008): $K_r = \tanh(a \cdot \xi^b)$, with coefficients for rock or smooth slopes; the reflected energy fraction is K_r^2.

6. Vertical (caisson) stability

The Goda (2000) pressures give the horizontal force P and uplift U on the caisson; sliding and overturning are then checked against the caisson's submerged weight W and base width B:

$SF_{slide} = \mu(W - U)/P$, $SF_{over} = (W \cdot B/2)/(M_P + M_U)$; both should exceed about 1.2.

7. Worked design summary (tool defaults)

Response	Result
Design conditions	H_s=3 m, T_p=8 s, h=8 m, slope 1:2, R_c=4 m, crest B=6 m
Front armour (rock, governing)	3.58 t, D_{n50}=1.11 m (Van der Meer)
Toe stone (N_{od}=2)	0.12 t, D_{n50}=0.35 m (N_s=5.4)
Run-up / overtopping (γ_f=0.40)	$R_{2\%}$=3.69 m, q=0.4 l/s/m

Transmission (R_c/H_s=1.3)	K_t=0.075 (negligible)
Reflection (rock)	K_r=0.29
Caisson (W=1800 kN/m)	H_{max}=5.40 m, P=477 kN/m, U=138 kN/m
Caisson stability	SF_{slide}=2.1, SF_{over}=2.0

References

CIRIA / CUR / CETMEF (2007). *The Rock Manual*, 2nd ed. C683, London.

EurOtop (2018). *Manual on Wave Overtopping of Sea Defences*, 2nd ed. Van der Meer, J. W. et al.

Goda, Y. (2000). *Random Seas and Design of Maritime Structures*, 2nd ed. World Scientific.

Van der Meer, J. W. (1988). Rock slopes and gravel beaches under wave attack. PhD thesis, Delft.

d'Angremond, K., van der Meer, J. W. and de Jong, R. J. (1996). Wave transmission at low-crested structures. *Proc. 25th ICCE*, ASCE.

Zanuttigh, B. and van der Meer, J. W. (2008). Wave reflection from coastal structures. *Coastal Engineering*, 55(10).

U.S. Army Corps of Engineers (1984, 2002). *Shore Protection Manual* / *Coastal Engineering Manual*.